Evidence for geomagnetic field excursions and secular variation from the Wrangell Volcanics of Alaska

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Paleomagnetic studies have been made on 36 late Tertiary lava flows (3–4 m.y.) from the Wrangell Volcanics. Final flow mean remanent magnetization directions show excursions of the geomagnetic field away from a mean corresponding to a geocentric axial dipole field. They also point to the possibility that such excursions may have been a more common occurrence at the time of extrusion of these lavas than appears to have been the case in Quaternary times. These excursions may be due to increased non-dipole field activity. Calculation of the paleosecular variation including field excursion data leads to high values of PSV which do not agree with existing models. Exclusion of field excursion data gives a result that is consistent with current PSV models, but does not allow differentiation between them.

Des études paléomagnétiques ont été faites sur trente-six coulées de lave du Tertiaire supérieur (3–4 Ma), provenant des roches volcaniques de Wrangell. Les directions du magnétisme rémanent moyen de la fin des coulées montrent des variations du champ géomagnétique qui s'éloignent d'une valeur moyenne correspondant à un champ dipolaire axial géocentrique. Elles évoquent aussi la possibilité que de telles variations peuvent avoir été beaucoup plus fréquentes pendant l'extrusion de ces laves que ce qui semble avoir été le cas pendant le Quaternaire. Ces variations peuvent être dues à une augmentation de l'activité non-dipolaire du champ. Le calcul de la variation paléoséculaire, en incluant les données sur les variations du champ, donne des valeurs élevées de la PSV, ce qui n'est pas en accord avec les modèles existants. L'exclusion des données sur les variations du champ donne un résultat qui est en accord avec les modèles connus de la PSV mais ne permet pas de les distinguer entre eux.

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Introduction

Knowledge of the temporal behavior of the geomagnetic field of the earth over time scales short compared to rates of polar wandering or continental drift is important in placing boundary conditions or limits on geomagnetic field models. Time-varying features of the geomagnetic field of interest that have been studied are reversals, intensities, secular variation, and field excursions.

In order to investigate these aspects of the geomagnetic field at a remote time in the geologic past it is necessary to sample the paleomagnetic field across a section of rock known to have been deposited and/or to have acquired a magnetic remanence more or less uniformly at a known rate. These conditions are rarely met in practice. Sections of sedimentary rocks without unconformities or sequences of lava flows are most commonly used for such studies.

This paper reports paleosecular variations (PSV) and geomagnetic field excursion (GFE) data from three sequences of lava flows in Alaska. The sampling area is in the Wrangell Mountains of south-central Alaska in rocks that were mapped as the Wrangell Volcanics of upper Tertiary age by Moffit *et al.* (1910) and Moffit (1943) (Fig. 1). The three different sections of flows were sampled within a few kilometres of each other near the abandoned mining town of Nabesna: Air II (9 flows), Castles (11 flows), and Wait Creek (16 flows) sections. Natural remanent magnetization (NRM) data from the Air II and Castles sections, as well as preliminary NRM results from 9 flows of the Wait Creek section, were published by Stone (1970). The data presented here represent small changes in the previously published data following alternating field (a.f.) demagnetization, and final results from all the Wait Creek flows.

The Castles and Air II sections were described by Stone (1970), who concluded at that time that both sections are approximately, but prob-

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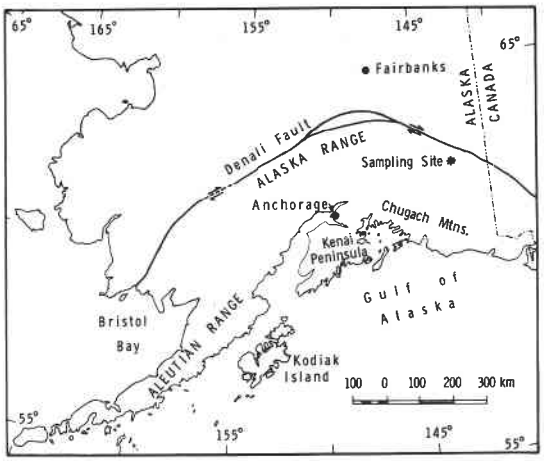


FIG. 1. Location map showing the general location of the Wrangell Volcanics and the area sampled.

ably not exactly contemporaneous with each other and with the Wait Creek section. Flows in both sections appear to be in their original attitude, which is approximately horizontal.

The section sampled at Wait Creek is about 500 m thick and has been described in detail by Deininger (1972). It consists of basaltic andesite and augite–hypersthene andesite lava flows, although the top-most flow sampled (flow 21) is more dacitic, being the lowest member of a series of dacite flows above the sampled section (Deininger 1972). The entire section dips about 10° to the west, this tilt being taken into account in the paleomagnetic data.

Age of the Sections Sampled

The age of all three sections samples is now considered to be about 3.4 m.y., as opposed to the original age quoted by Bingham (1971) of 6.5 m.y. This change is due to several new K–Ar age determinations made by J. Deininger and D. L. Turner (personal communication 1974) for both the Wait Creek and Castles sections, and also for a number of nearby and apparently contemporaneous sequences of flows. The original, and much higher age quoted was determined from a whole-rock sample from the lower part of the Wait Creek section, which on further inspection was found to contain significant quantities of glass. This glass may well account for the apparently anomalous ages found.

Geologic relationships between the sites at Wait Creek and the Castles point to the Castles being a little higher stratigraphically. This relationship is based largely on the tracing and extrapolation of the inter-flow ash layers mapped by Deininger (1972). The sections sampled at Air II, based exclusively on the relative elevations of the sites, would be intermediate in age. It is not thought, however, that there can be any large age difference between any of them.

Methods and Results

Essentially the same methods were used in collecting samples and processing data from all three sections.

Sampling was carried out using a field coring drill (Doell and Cox 1967) with which 3 to 7 cores per flow were drilled (because of difficulty in drilling, three flows in the Castles section are represented by less than 3 cores). Cores were oriented using a sun compass wherever possible and by sighting on geographic features whose true bearing had been determined using the sun and topographic maps.

Measurements were done on a 5 Hz spinner magnetometer (Foster 1966). Alternating field demagnetization studies were carried out using magnetic cleaning apparatus similar to that described by McElhinny (1966).

Detailed stepwise a.f. cleaning up to more than 2000 Oe was done on 24 pilot specimens from 13 of the Wait Creek flows. All specimens showed good end-points below 380 Oe peak a.f.; the relative change in intensity at this field, J/J_0, was between 0.3 and 0.8 (Fig. 2). It was therefore decided to apply blanket treatment of 380 Oe to one specimen from each of the other cores in the section. This includes flows 6 through 14, for which the NRM data were reported by Stone (1970) (there numbered 1 through 9). All further analysis is confined to these cleaned directions. Flow mean directions based on a.f. cleaned data and corrected for attitude of the section are listed in Table 1 and are shown in Fig. 2.

Since fewer samples were collected from the Castles and Air II sections, one specimen from each sample was stepwise a.f. demagnetized until an end point was reached (maximum treatment used was 340 Oe (peak field)). The final directions were then chosen from a.f. cleaned data following the minimum dispersion method (Irving *et al.* 1961), though in nearly all cases movement is slight and directions and dispersions do not differ significantly from treatment

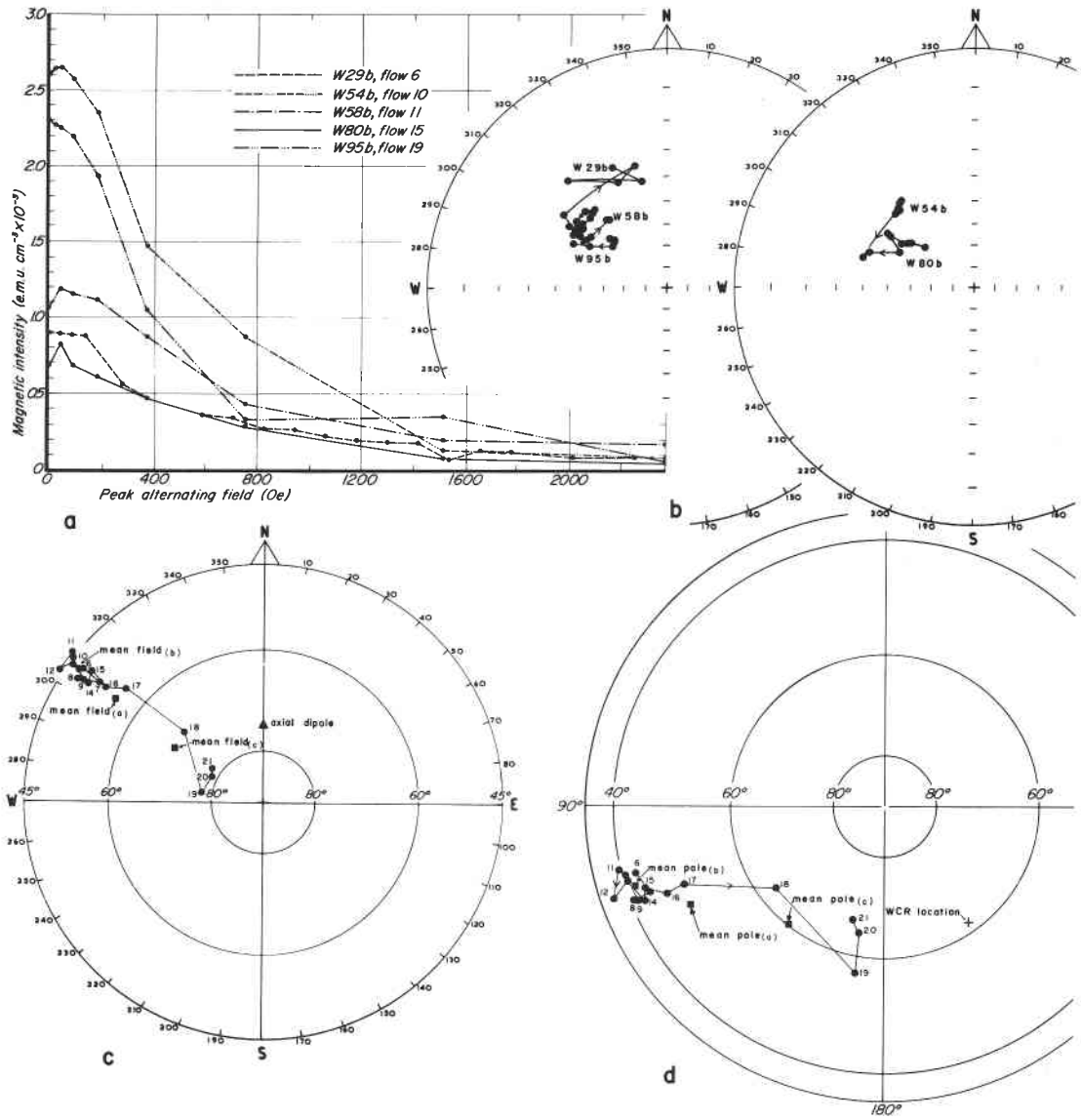


FIG. 2. Paleomagnetic data for Wait Creek section. (a) Magnetic intensity and (b) directions of typical pilot specimens during a.f. demagnetization (arrows point toward higher peak fields). (c) Flow mean directions, and (d) corresponding poles. Flow numbers are indicated. Section means calculated as described in text and Table 2. Flow means from Table 1. All vectors plotted on lower hemisphere using Wulff stereonet.

to treatment. For the two flows where $N = 1$, the direction that changed least with successive demagnetization was selected.

The results for the Castles and Air II sections are listed in Table 1 and Fig. 3. The data differ only slightly from the NRM results reported by Stone (1970).

Discussion

The 36 flows in the three sections are normally magnetized and the within-flow dispersions are generally small, with δ, as defined by Wilson (1959), being less than 10° for all but 3 flows (Table 1). Only 4 flows (W18, A02, A04, and A05) have α_{95} circles of confidence that include

TABLE 1. Flow mean paleomagnetic data for the Wrangell Volcanics, Alaska

Flow No.	Treatment	N	k	δ	α_{95}	D	I	λ	ϕ
Wait Creek (143°6.1′W, 62°19.1′N)									
W21	380 Oe	5	417.8	3.6	3.7	302.9	78.8	66.4	166.2
W20	380	6	71.0	8.8	8.0	295.7	79.2	64.1	169.6
W19	380	6	28.2	14.0	12.8	291.5	74.6	59.0	155.9
W18	380	6	12.5	21.0	19.7	304.0	77.5	66.1	160.4
W17	380	6	41.4	11.5	10.5	302.5	58.2	48.6	120.1
W16	380	7	108.0	7.2	5.8	306.2	53.0	46.0	112.0
W15	380	6	311.2	4.2	3.8	307.0	48.9	43.0	108.5
W14	380	6	62.1	9.4	8.6	304.5	49.7	42.5	111.5
W13	380	4	225.1	4.7	6.1	306.3	45.8	40.5	107.3
W12	380	5	240.4	4.7	4.9	303.3	43.5	37.6	109.2
W11	380	6	122.5	6.7	6.1	308.3	43.5	39.8	104.1
W10	380	6	214.4	5.1	4.6	307.9	44.5	40.3	105.1
W09	380	5	587.7	3.0	3.2	304.1	48.1	41.2	110.9
W08	380	6	804.6	2.6	2.4	303.9	47.0	40.3	110.4
W07	380	6	69.3	8.9	8.1	306.3	50.5	43.9	110.1
W06	380	6	333.6	4.1	3.7	306.6	46.9	42.3	105.6
Castles (143°2.1′W, 62°24.4′N)									
C11	127 Oe	1	–	–	–	173.5	78.4	40.1	220.1
C10	85	5	123.4	6.5	6.9	167.7	84.2	51.0	220.8
C09	42	5	643.1	2.9	3.0	150.2	76.8	39.1	232.7
C08	85	2	5489.9	0.8	3.4	136.4	79.4	45.3	237.0
C07	127	4	342.3	3.8	5.0	152.5	80.5	45.1	228.9
C06	NRM	4	180.5	5.2	6.9	108.8	79.1	50.6	249.3
C05	85	4	1471.3	1.8	2.4	92.5	83.2	58.9	243.6
C04	255	5	310.1	4.1	4.4	111.5	80.7	52.0	245.0
C03	212	6	391.3	3.7	3.4	102.2	80.8	54.3	248.0
C02	85	4	78.5	7.9	10.4	132.1	83.3	52.2	233.0
C01	212	1	–	–	–	117.1	74.0	41.5	253.2
Air II (143°10.7′W, 62°55.4′N)									
A09	NRM	5	359.1	3.8	4.0	340.2	76.8	80.9	151.9
A08	85 Oe	4	635.1	2.8	3.6	345.5	78.0	82.3	170.2
A07	85	3	753.4	2.4	4.5	341.7	80.8	78.0	189.1
A06	42	5	1346.2	2.0	2.1	333.5	79.6	77.0	173.6
A05	NRM	4	277.5	4.2	5.5	347.8	75.2	84.3	130.8
A04	42	3	69.9	7.9	14.9	007.5	63.0	71.6	019.7
A03	85	4	957.6	2.3	3.0	359.0	70.0	81.6	040.9
A02	42	4	166.4	7.0	9.2	003.8	79.8	81.9	226.1
A01	NRM	4	1775.7	1.7	2.2	355.9	78.1	84.8	198.9

Within each section flows are listed in stratigraphic order (youngest first); treatment is peak a.f. washing used; N = number of cores; k = precision parameter (Fisher 1953); $\delta = \cos^{-1} R/N$ where R = length of resultant of sum of N unit vectors (Wilson 1959); α_{95} = 95% confidence limit (Fisher 1953 ; D, I = mean declination, inclination; λ, ϕ = North latitude and East longitude of pole position.

the present and/or axial field directions. These facts together with small movements under a.f. demagnetization already noted, indicate that the directions measured probably represent primary directions of magnetization.

Field Excursions

Flows W06 through W16 of the Wait Creek section are noteworthy in that they cluster closely about a mean that is more than 35° from the axial dipole or present field directions and corresponds to a pole in Inner Mongolia (entry 1b, Table 2, and Fig. 2). These flows evidently were extruded during a type of geomagnetic event that has come to be known as a geomagnetic field excursion (*e.g.*, Doell and Cox 1972; Barbetti and McElhinny 1972).

The entire Castles section also appear to have recorded a GFE, though a different one from that found at Wait Creek. These 11 flows are grouped about a direction having a pole in northwestern Washington State (entry 2, Table 2).

The remaining flows give evidence of more typical field behavior. Wait Creek flows W17 to W21 show a systematic return to a direction about 10° from the axial dipole field (Fig. 2), while the nine Air II flows are grouped about a mean only

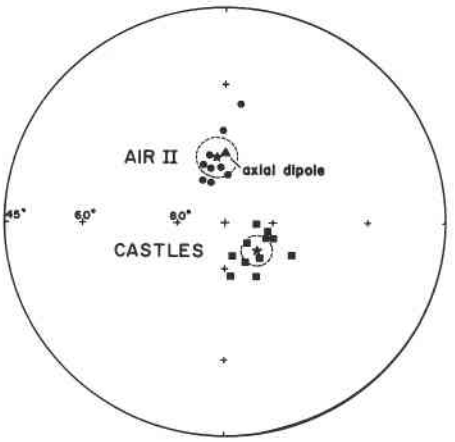


FIG. 3. Paleomagnetic flow mean directions for Air II (circles) and Castles (squares) with section means (stars) and α_{95} circles of confidence. All points plotted on lower hemisphere using Wulff stereonet.

2° from the axial dipole field (entry 3, Table 2; Fig. 3).

Field observations of flow attitude preclude tilting as an explanation for the divergent directions, and the return toward the dipole field direction exhibited by flows W17 to W21 provides good evidence that the observed GFE's are not due to tectonic causes.

The excursions of the geomagnetic field recorded in two of the three sections sampled are too large, on current models, to be due to PSV alone. In terms of flows, 27 out of 36 flows have recorded excursions. Assuming that the lavas are roughly uniformly spaced in time (a dubious assumption at best) or alternatively that the three sections sampled represent different times selected randomly (probably a good assumption), it appears that field excursions occurred about two-thirds of the time during the period sampled. In contrast, Doell and Cox (1972) found only about 4% of 1035 lavas from widely scattered locations, with ages less than 5–10 m.y., gave paleolatitudes of less than 50°. Using their figure of 4% to represent the excursion ratio of an infinite population of lava flows, one may calculate the probability of drawing a sample having the excursion ratios found for the Wrangell Volcanics. For either of the two sampling assumptions described above, this probability is less than 0.5%. This leads to two possible conclusions. Either the number of field excursions occurring in the time period sampled was very high (or perhaps very local), or the results were due to fortuitous sampling, the probability of which is very low. The latter case is, however, intuitively more appealing.

Since the age of the Wrangell Volcanics is near the Gilbert Reversed-Gauss Normal Epoch boundary, the geomagnetic field behavior recorded may have occurred during polarity transitions. Because of the stratigraphic relationships between the sections sampled, two different polarity transitions are required: The Wait Creek section would have sampled the 'tail end' of a reversed–normal transition while the stratigraphically higher Castles section would represent either a time near the beginning of a normal–reversed transition or a time near the end of a reversed–normal transition. No rocks of reversed polarity have been sampled in the Wrangell Volcanics. This fact, coupled with the low probability of fortuitously sampling two separate reversals at the same point in their development or demise, makes it unlikely that polarity transitions rather than GFE's have been sampled.

Paleosecular Variation

To obtain a reliable measurement of PSV it is necessary to be assured that the sequence of paleomagnetic data points covers both a long enough time interval to include the longest period of secular variation and be of sufficient number and evenly enough distributed throughout that interval to have recorded the entire PSV spectrum. This assurance is never completely achieved in practice. Since radiometric dating does not have the required precision, most workers use various geomorphological clues to argue that the time interval represented is greater than about 10^4 years and that the lava flows are more or less evenly spaced in time (*e.g.*, Doell 1969; Stone 1970). The minimum useful number of independent flows is generally taken to be 10 (Bingham and Stone 1972; Creer and Sanver 1970; Brock 1971).

In Table 2 statistical parameters are defined and calculated for several combinations of flows from the three sections. Calculations have been carried out both with respect to magnetic vector directions and the locations of the equivalent virtual geomagnetic pole (VGP), to facilitate comparison with other PSV studies. In calculating K_F, the precision parameter reflecting ancient variations of the geomagnetic field, we have used Doell's (1970) value $K_A = 2100$ for the precision parameter due to magnetic anomalies

TABLE 2. Wrangell Volcanics. Statistical analysis of flow mean paleomagnetic data

	Analysis of direction														
	B	D	I	R	α_{95}	k_T	k_W	k_F	s_F	s_I	s_μ	λ	ϕ	dp	dm
1a. Wait Creek (a)	16	304.5	55.5	15.5736	6.3	35.2	79.1	38.6	13.0	10.5	17.1	47.5	115.1	6.4	9.0
1b. Wait Creek (b)	11	305.8	47.4	10.9842	1.8	613.6	194.4	–	–	–	–	41.5	108.5	1.5	2.3
1c. Wait Creek (c)	6	301.8	69.4	5.8635	11.2	36.6	35.6	44.9	12.1	8.9	19.1	58.8	137.1	16.3	19.1
2. Castles	11	131.3	80.9	10.9574	3.0	234.8	277.8	316.3	4.6	3.6	6.4	48.6	237.2	5.6	5.8
3. Air II	9	353.0	76.0	8.9496	4.1	158.9	310.1	184.4	5.9	4.5	8.5	86.6	150.6	7.0	7.6
4. Lines 1a, 2, 3	36	314.4	75.0	33.4839	6.6	13.9	135.1	14.2	21.5	18.5	25.6	69.1	146.0	11.0	12.0
5. Lines 1c, 2, 3	26	341.1	84.9	25.0912	5.5	27.5	120.5	29.1	15.0	12.6	18.5	71.6	206.5	10.7	10.9
6. Lines 3 and W17 through W21	14	331.6	76.9	13.7952	5.2	63.5	82.9	76.9	9.2	7.4	12.3				

	Analysis of poles									
	B	R	A_{95}	K_T	K_N	K_F	S_F	S_I	S_μ	λ	ϕ
1a. Wait Creek (a)	16	15.3453	7.9	22.9	55.8	25.0	16.2	13.1	21.3	49.3	117.5
1b. Wait Creek (b)	11	10.9858	1.7	703.6	186.7	–	–	–	–	41.6	108.6
1c. Wait Creek (c)	6	5.7687	14.7	21.6	14.6	29.4	14.9	10.9	23.5	60.1	141.2
2. Castles	11	10.8498	5.6	66.6	78.0	90.1	8.5	6.6	11.8	48.7	237.0
3. Air II	9	8.8697	6.6	61.4	97.2	75.5	9.3	7.1	13.4	86.2	156.5
4. Lines 1a, 2, 3	36	30.1941	10.6	6.0	43.9	6.2	32.5	28.0	38.7	71.5	161.0
5. Lines 1c, 2, 3	26	23.3353	9.8	9.4	31.3	10.2	25.4	21.4	31.3	70.6	210.7
6. Lines 3 and W17 through W21	14	13.4960	8.0	25.8	25.7	33.4	14.0	11.2	18.7	77.7	152.4

(a) = flows W06–W21, (b) = flows W06–W16, (c) = flows W17–W21 + unit weight to flows W06–W16; B = number of flows; D, I = mean declination, inclination; R = length of resultant of B unit vectors; α_{95}, A_{95} = 95% confidence limits (Fisher 1953); k_T, $K_T = B - 1/B - R$ = total observed between-flow precision parameter (Fisher 1953); $k_W = (81/\delta_{rms})^2$ = within-flow precision parameter for field directions where $\delta_{rms} = \sqrt{\Sigma\delta^2/B - 1}$; $K_W = C^2 k_W$ where $C = \frac{1}{2}(1 + 3\cos^2 I + \sqrt{1 + 3\cos^2 I}$ (Cox 1962); k_F, K_F are precision parameters reflecting ancient field variations, using $1/k_F = 1/k_T - 1/Nk_W - 1/k_A$, where N is mean number of samples per flow and $C^2 k_A = K_A = 2100$ = precision parameter for effect of magnetic anomaly existing at time of eruption (Doell 1970); s_F, $S_F = 81/\sqrt{k_F}$, $81/\sqrt{K_F}$ = dispersion due to ancient field variations; s_I, S_I and s_μ, S_μ are lower and upper 95% confidence limits on s_F, S_F (Cox 1969); λ, ϕ are north latitude, east longitude of VGP; dp, dm are semi-axes of 95% oval confidence about VGP.

at the time of extrusion. While this value was derived for a volcanic island instead of an inland volcano, its effect on the computation is small and any error in K_A is not likely to be significant to the final result.

In his preliminary report on the Wrangell Volcanics, Stone (1970) concluded that because the within-flow scatter was larger than the between-flow scatter it was not possible to obtain a measure of PSV in any of the three sections. In both the Castles and Air II sections, it is clear that using a.f. cleaned data has improved the within-site statistic (k_W) to the point where $k_W > k_T$, the between-flow statistic (entries 2 and 3, Table 2). Though the difference is not large and the numbers of flows only barely sufficient, it seems that values of k_F (or $s_F = 81\sqrt{k_F}$) are probably real measures of field changes during the time of extrusion of these two sections.

As already noted, flows W06 through W16 in the Wait Creek section defining a field excursion are very tightly grouped. In fact, the observed between-flow precision is so much larger than the within-flow precision parameter (line 1b, Table 2) that these flows almost certainly represent either a very short period of time or a period of high field stability. In the former case, it is reasonable to combine all 11 flows together (line 1b, Table 2) and give unit weight to the mean in combining with other flows (lines 1c and 5, Table 2). In the latter case, that of high field stability, giving unit weight to all flows is appropriate (lines 1a and 4, Table 2). In this case $k_W > k_T$, indicating that real field variations are represented by k_F or s_F. In the case where the 11 flows of the field excursion are together given unit weight, the between-flow scatter is slightly tighter than the within-flow scatter. However, the difference is not significant. In addition, the small number of 'flows' (B = 6) in this case makes this value less desirable.

Since two of the three sections have evidently sampled GFE's, it is not clear which, if any, of the dispersion values discussed thus far best represents the PSV for the time of extrusion of the Wrangell Volcanics. If it is supposed that field excursions represent a longer period extension of the PSV spectrum, then GFE data should be included along with data not representing field excursions. Alternately, if field excursions and PSV are regarded as essentially different phenomena, then GFE and non-GFE data should not be combined, though GFE data may be used alone.

Without for the moment combining data from different sections, there are now four possible dispersion parameters (*cf.* lines 1a, 1c, 2, 3 of Table 2). Both the Wait Creek values combine directions representing excursions of the paleomagnetic field with directions not related to field excursions, and hence are larger (s_F = 12.1 and 13.0°), while the Castles and Air II values comprise either entirely GFE data or entirely non-GFE data and therefore are smaller (s_F = 4.6 and 5.9°).

In order to be assured of covering a sufficiently long time interval, it is appropriate to combine data from the different sections. In the case where PSV and GFE's are considered to represent different manifestations of the same sort of physical process we may combine the data in the manner indicated in lines 4 and 5 of Table 2 giving s_F = 21.5 and 15.0°, respectively. Under the assumption that GFE's and PSV should not be combined, the method of combination given in line 6, Table 2 is appropriate to PSV; this gives s_F = 9.2°.

To facilitate comparison of these results with other work, all of the statistical calculations have been carried out in reference to VGP's as well as to directions (Table 2). Referred to poles the two assumptions discussed above yield S_F = 32.5 or 25.4° (lines 4 and 5, Table 2) if GFE's are included in PSV, and S_F = 14.0° (line 6) if GFE's are excluded. The former values are clearly too high to fit any of the commonly discussed secular variation models (*e.g.*, Irving and Ward 1964; Creer *et al.* 1959; Cox 1962, 1970), which for the latitude of the Wrangell Volcanics (62°) require S_F in the approximate range 11 to 18°; even the 95% confidence limits on S_F (Cox 1969) do not fall in this range. The value that excludes the effects of GFE's, S_F = 14°, almost exactly includes this range within its 95% confidence limits. Therefore, it is not possible on the basis of this result to differentiate between the various secular variation models.

Concluding Remarks

The paleomagnetic results from the Wrangell lava flows point to two geomagnetic field excursions occurring at about 3.4 m.y. The probability of sampling three separate sections and finding that two of them contain field excursions is very

low if the frequency of occurrence of such excursions is that calculated by Doell and Cox (1972). This leads to the conclusion that the rate of occurrence of such events must have been considerably higher during this period. However, this conclusion has to be tempered by the small number of sites compared. It is unfortunate that logistic problems have precluded further sampling, most other suitable sections being somewhat inaccessible.

Because of the observed excursions, and also because of the unknown time spans for each of the sequences of flows, the data have been combined in a number of ways to investigate the paleosecular variation of the field. Including the excursion data leads to values of PSV higher than expected for this latitude, while excluding it makes the number of data points available marginal but gives values consistent with current paleosecular variation models.

Acknowledgments

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BARTETTI, M. and MCELHINNY, M. 1972. Evidence of a geomagnetic excursion 30 000 yr BP. Nature, **239**, pp. 327–330.

BINGHAM, D. K. 1971. Paleosecular variation of the geomagnetic field in Alaska. Unpubl. PhD diss., Univ. Alaska, Fairbanks, Alaska, 143 p.

BINGHAM, D. K. and STONE, D. B. 1972. Secular variation in the Pacific Ocean region. Geophys. J. R. Astron. Soc. **28**, pp. 337–355.

BROCK, A. 1971. An experimental study of paleosecular variation. Geophys. J. R. Astron. Soc. **24**, pp. 303–317.

COX, A. 1962. Analysis of present geomagnetic field for comparison with paleomagnetic results. J. Geomag. Geoelect. **13**, pp. 101–112.

——— 1969. Confidence limits for the precision parameter κ. Geophys. J. R. Astron. Soc. **17**, pp. 545–549.

——— 1970. Latitude dependence of the angular dispersion of the geomagnetic field. Geophys. J. R. Astron. Soc. **20**, pp. 253–269.

CREER, K. M., IRVING, E., and NAIRN, A. E. M. 1959. The paleomagnetism of the Great Whin Sill. Geophys. J. R. Astron. Soc. **2**, pp. 306–323.

CREER, K. M. and SANVER, M. 1970. Secular variations of the geomagnetic field in the Quaternary and Late Tertiary. *In* Paleogeophysics (S. K. Runcorn, *Ed.*). Academic Press, London, pp. 91–100.

DEININGER, J. 1972. Petrology of the Wrangell Volcanics near Nabesna. Unpubl. MSc thesis, Univ Alaska, Fairbanks, Alaska, 66 p.

DOELL, R. R. 1969. Paleomagnetism of the Kau Volcanic Series, Hawaii. J. Geophys. Res. **74**, pp. 4857–4868.

——— 1970. Paleomagnetic secular variation study of lavas from the Massif Central, France. Earth Planet. Sci. Lett. **8**, pp. 352–362.

DOELL, R. R. and COX, A. 1967. Paleomagnetic sampling with a portable coring drill. *In* Methods in paleomagnetism (D. W. Collinson, K. M. Creer, and S. K. Runcorn, *Eds.*). Elsevier, Amsterdam, Netherlands, pp. 21–25.

——— 1972. The Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle. *In* The nature of the solid earth (E. C. Robertson, *Ed.*). McGraw-Hill, New York, N.Y., pp. 245–284.

FISHER, SIR RONALD. 1953. Dispersion on a sphere. Proc. R. Soc. **A 217**, pp. 295–305.

FOSTER, J. H. 1966. A paleomagnetic spinner magnetometer using a fluxgate gradiometer. Earth Planet. Sci. Lett. **1**, pp. 463–466.

IRVING, E., STOTT, P. M., and WARD, M. A. 1961. Demagnetization of igneous rocks by alternating magnetic fields. Phil. Mag. **6**, pp. 225–241.

IRVING, E., and WARD, M. A. 1964. A statistical model of the geomagnetic field. Pure Appl. Geophys. **57**, pp. 47–52.

MCELHINNY, M. W. 1966. An improved method for demagnetizing rocks in alternating magnetic fields. Geophys. J. R. Astron. Soc. **10**, pp. 369–374.

MOFFIT, F. H. 1943. Geology of the Nutzotin Mountains, Alaska. U.S. Geol. Surv. Bull. 933-B, pp. 103–199.

MOFFIT, F. H., KNOPF, A., and CAPPS, S. R. 1910. Mineral resources of the Nabesna–White River district, Alaska. U.S. Geol. Surv. Bull. 417.

STONE, D. B. 1970. Paleosecular variation in Alaska. *In* Palaeogeophysics (S. K. Runcorn, *Ed.*). Academic Press, London, pp. 117–126.

WILSON, R. L. 1959. Remanent magnetism of late Secondary and early Tertiary British rocks. Phil. Mag. **4**, pp. 750–755.